

Assignment

Large File Processing & Search

Problem Statement

Build a backend service that allows users to upload and search through **large text files**.

The service will run on a machine with **4 GB of RAM**, while individual files can be as large as **10 GB**.

Once a file has been uploaded, its contents should become searchable through an API.

Requirements

The service should allow a client to:

- Upload a file of up to **10 GB**.
- Resume an upload if the connection is interrupted.
- Check the status of an upload and its processing.
- Search the contents of an uploaded file.
- Retrieve the relevant sections of the file for a search query.

The system should:

- Work within the available **4 GB RAM**.
- Support multiple uploads.
- Handle failures gracefully.
- Persist relevant file and processing metadata.
- Process the uploaded file so that it can be searched efficiently.

Search

Allow users to search using natural language and return the most relevant sections of the file based on meaning rather than exact keyword matches.

For example, if the file contains:

Connection to database failed after 30 seconds.

A search for:

database connectivity problems

should be able to return the relevant section.

The implementation approach is left to you.

Technical Requirements

- Use **Python, Go**, or equivalent.
- Use any suitable backend framework and database.
- You may choose the search/indexing/vector database technology.
- The application should be runnable locally.
- Provide API documentation.
- Automated tests are good to have.
- Docker image is optional.

Design Discussion

Include a short section in the README explaining:

1. How you handle a **10 GB file on a 4 GB RAM machine**.
2. How you handle interrupted uploads.
3. How you would support multiple concurrent uploads.
4. How you would process and index a large file efficiently.
5. How your semantic search works.
6. What you would change if the service needed to support **thousands of concurrent uploads and searches**.

Use of AI Coding Tools

You are free to use **AI-assisted coding and code-generation tools** during the assignment. However, you are expected to **understand and be able to explain all parts of your implementation and the technical decisions made**.

Please mention which tools you used and briefly explain how you validated or modified the generated code.

Submission

Please provide the public github repository with:

- Source code
- README with setup instructions and design decisions
- API documentation
- Any configuration or Docker files required to run the application

NOTE: The assignment does not need to be production-ready. We are primarily interested in your **engineering approach, implementation quality, and reasoning behind your design decisions.**